SMARTCAMPUS AI

Enterprise-Grade Autonomous Smart Campus & Multi-Evidence Biometric Attendance Platform

Course:	Bachelor of Engineering (B.E.) in Computer Science & Engineering
Project Phase:	Major Project Phase-II (Course Code: BCS786)
Institution:	Vemana Institute of Technology, Bangalore
Department:	Department of Computer Science and Engineering
Project Architecture:	Next.js 16 + NestJS 10 Gateway + FastAPI AI Vision Engine + FAISS
Document Type:	Exhaustive Step-by-Step Weekly Engineering Report & Complete System Audit
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Executive Summary: SmartCampus AI is an advanced, privacy-first computer vision and institutional automation platform. It automates multi-student classroom face recognition, contactless attendance logging, timetabling, behavior diagnostics, and edge IoT monitoring. Biometric templates are preserved as 512-D L2-normalized ArcFace feature vectors indexed within local FAISS vector storage, completely eliminating cloud dependencies for sensitive data. The system implements strict multi-evidence identity consolidation and automated cluster purity validation, enforcing the invariant: **1 Physical Person = 1 Identity Candidate**.

System Architecture & Table of Contents

This document provides a comprehensive, weekly chronological breakdown of the design, mathematical formulations, implementation milestones, algorithmic optimizations, and verification procedures of the entire SmartCampus AI ecosystem from inception to enterprise readiness.

Section / Module	Focus & Key Deliverables
Week 1	Monorepo Architecture, PostgreSQL Schema, Prisma ORM, Multi-Tenant Auth Gateway
Week 2	SCRFD 10G Detection, ArcFace 512D Embeddings, FAISS Vector Engine, Passive Attendance
Week 3	Back-Row Small-Face Tile Upscaling, MediaPipe Pose/Quality Gate, Diagnostics Dashboard
Week 4	Zero-Regression IoU Temporal Tracker, Occlusion Handling, Track Recovery
Week 5	NestJS Core Services (Students, Attendance, IoT Devices, Classrooms, Risk Profiling)
Week 6	Next.js 16 Web Dashboard, Glassmorphic UI, Live RTSP Stream Overlays, WebSocket IPC
Week 7	Classroom Video Enrollment Studio, Two-Stage Pipeline, Crop Resolution Engine
Week 8	Track-Level Face Profiles, HDBSCAN Density Clustering, Over-Clustering Prevention
Week 9	Biometric Data Governance, Database Reset Flow, FAISS Vector Index Synchronization
Week 10	Multi-Evidence Identity Consolidation (Strict Cross-Matrix Decision & Anti-Chaining)
Week 11	Internal Cluster Purity Validation, Graph Connected-Component Auto-Splitting
Week 12	Comprehensive Unit Test Suite (12/12 Tests Passing), Production Build, System Verification

Week 1: Monorepo Architecture, PostgreSQL Schema & Auth Gateway

Primary Objective: Establish the foundational monorepo workspace, define the institutional multi-tenant database schema with Prisma ORM, and build the NestJS secure authentication gateway.

Detailed Work Done & Engineering Achievements:

- **Turborepo Monorepo Setup:** Configured apps/web (Next.js 16), apps/api (NestJS 10), and apps/ai-service (FastAPI) with shared environment variable synchronization and npm workspaces.
- **PostgreSQL Database Modeling (Prisma ORM):** Authored apps/api/prisma/schema.prisma supporting multi-tenant institutional isolation: Institution → Department → Classroom → Course / Subject → User / Role → Student / Teacher.
- **JWT & RBAC Security Layer:** Built NestJS Passport JWT module supporting short-lived access tokens (15-min) and HTTP-only refresh tokens (7-day). Implemented role guards for SUPER_ADMIN, INSTITUTION_ADMIN, HOD, TEACHER, and STUDENT.
- **Local Infrastructure Orchestration:** Configured docker-compose.yml for PostgreSQL 16, Redis 7 (caching/queues), Mosquitto MQTT (IoT messaging), and local directory mounts.
- **Webcam Simulator Bypass:** Implemented MockVideoCapture in camera_service.py to allow seamless headless testing on servers or machines without physical cameras.

Deliverables: *apps/api/prisma/schema.prisma, src/auth/, docker-compose.yml, start.sh*

Effort: *42.0 Hours* | ***Verification:*** *PostgreSQL migrations applied cleanly, JWT token issuance verified.*

Week 2: Deep Biometric Pipeline, ArcFace 512D & FAISS Vector Search

Primary Objective: Construct the high-accuracy computer vision pipeline in Python FastAPI, integrating deep facial detection, ArcFace feature extraction, and sub-millisecond vector retrieval.

Detailed Work Done & Engineering Achievements:

- **SCRFD 10G Face Detection:** Integrated InsightFace SCRFD (Sample and Computation Redistribution for Face Detection) 10G model to detect multi-scale faces with 5-point facial keypoints.
- **ArcFace ResNet50 Feature Extractor:** Built embedding_service.py using ONNX Runtime. Extracted 512-dimensional facial representations and enforced strict L2 unit-norm normalization (||v|| = 1.0).
- **Sub-Millisecond FAISS Vector Indexing:** Built faiss_service.py implementing IndexFlatIP (Inner Product for Cosine Similarity). Allowed querying thousands of enrolled students in under 0.8 ms.
- **Unsupervised Face Clustering:** Integrated initial HDBSCAN density clustering (hdbscan_clustering.py) to automatically group unassigned facial crops captured across video frames.
- **Passive Attendance State Machine:** Implemented attendance_engine.py requiring persistent temporal voting across consecutive frames to confirm student presence.

Mathematical Formula: *ArcFace Cosine Similarity: $S_C(u, v) = u \cdot v$ where $||u|| = ||v|| = 1.0$*

Deliverables: *embedding_service.py, faiss_service.py, attendance_engine.py, vectors/face_index.faiss*

Week 3: Back-Row Small-Face Recovery & MediaPipe Quality Gates

Primary Objective: Resolve small-face detection drops for back-row classroom seating and eliminate false matches caused by extreme head pose angles or motion blur.

Detailed Work Done & Engineering Achievements:

- **Adaptive 1.8x ROI Tile Upscaling:** Implemented an automated upper-classroom tile upscaling pass in classroom_pipeline.py (targeting frame[0:650, :]) to recover small faces (20-35 px) with coordinate re-projection and Non-Maximum Suppression (IoU 0.40).
- **MediaPipe 3D Pose Estimation:** Built mediapipe_face_quality.py to calculate 3D head yaw, pitch, and roll. Samples exceeding 35° yaw or 25° pitch are rejected before vector generation.
- **Laplacian Variance Blur Filtering:** Evaluated image sharpness using Laplacian variance; rejected blurred frames (score < 25.0) to maintain high embedding quality.
- **Real-Time Diagnostic Telemetry:** Created a live diagnostics dashboard with 22 tracking metrics: Processing FPS, SCRFD Detections, MediaPipe Quality Passes, and ArcFace Embedding Counts.

Deliverables: *mediapipe_face_quality.py, classroom_pipeline.py, Diagnostic Telemetry API*

Performance Benchmark: *Back-row face recall increased from 68% to 94.5% with < 180ms latency per frame.*

Week 4: Zero-Regression IoU Temporal Tracker & Occlusion Handling

Primary Objective: Build a stable, high-speed multi-face tracker that maintains identity track continuity across temporary occlusions and student head turns without modifying locked tracker algorithms.

- **IoUTracker Algorithm (face_tracker.py):** Implemented spatial Intersection-over-Union tracking with motion vector estimation, maintaining active track continuity across up to 90 lost frames (45 seconds at 2 FPS).
- **Anti-Switching Logic:** Preserved distinct track IDs for adjacent students in dense classroom seating, preventing ID swaps during student movement.
- **Temporal Representation Buffer:** Each track maintains up to 15 highest-quality face crops and corresponding 512D embeddings for deferred downstream clustering.

Week 5: Enterprise NestJS Core Services & Academic Workflows

Primary Objective: Build comprehensive backend business logic for student rosters, academic departments, classroom timetable management, attendance corrections, and IoT camera management.

- **Students Management Module (src/students/):** CRUD operations, USN/Roll Number indexing, student photo galleries, and automated CSV/Excel batch import.
- **Attendance Controller & Engine (src/attendance/):** Real-time session triggers, manual attendance override auditing, student risk profiling (absenteeism alerts), and attendance percentage aggregation.
- **IoT & Device Management (src/iot/, src/devices/):** Camera heartbeat tracking, RTSP URL provisioning, and remote device command dispatch.
- **Academic Scheduling:** Timetable management linking Classrooms, Subjects, Teachers, and active ClassSessions.

Week 6: Next.js 16 Web Dashboard & RTSP Live Monitoring

Primary Objective: Build a responsive, glassmorphic administrative interface in Next.js 16 with real-time RTSP stream viewing and WebSocket progress synchronization.

- **Modern UI/UX Design System:** Implemented dark-mode glassmorphic aesthetics with Tailwind CSS, Lucide icons, and Outfit typography.
- **Live Monitoring Hub (/dashboard/live-monitoring):** Multi-camera grid with real-time bounding box overlays: Green (Active/Tracked), Blue (Predicted), Yellow (Lost).
- **WebSocket Progress Integration:** Real-time progress bars for multi-camera video analysis broadcasting frame count, detection rate, and cluster count.

Week 7: Face Enrollment Studio & Crop Resolution Engine

Primary Objective: Build the contactless classroom enrollment interface and eliminate runtime crashes caused by malformed crop URLs.

- **Two-Stage Processing Architecture:** Stage 1 processes raw video frames with IoU tracking; Stage 2 extracts ArcFace feature vectors in deferred batch mode.
- **Safe Crop URL Resolver:** Developed resolveCropUrl() in face-enrollment/page.tsx to safely parse complex crop objects, relative storage paths, and null states without throwing gUrl.startsWith is not a function.
- **Timeout & Hang Protection:** Added AbortController with 4s timeout on cluster fetching, preventing infinite loading spinners.

Week 8: Track-Level Aggregation & Over-Clustering Elimination

The Problem: Initial naive clustering created one cluster per detection box or temporary track, resulting in 54+ identity candidates for a classroom of only 25 physical students.

Engineering Resolution:

- **Track-Level ArcFace Profiles:** Aggregated all sample embeddings inside a track into a single L2-normalized track centroid vector C_t = mean(v_i) / ||mean(v_i)||.
- **HDBSCAN Cosine Distance Matrix:** Fed track-level centroids into HDBSCAN with precomputed cosine distance, clustering dense tracks together and separating transient noise tracks.
- **Medoid Representation:** Calculated the medoid embedding (the member sample with maximum cosine similarity to the centroid) to represent the cluster in downstream FAISS searches.

Week 9: Biometric Data Governance & Synchronization Reset

Primary Objective: Implement an enterprise biometric reset workflow to allow testing fresh classroom sessions with zero stale vector artifacts.

- **Backend Reset API (POST /api/students/reset-all-enrollment):** Clears all PostgreSQL biometric fields (faceEnrolled = false, faceEmbedding = null, facePhotoUrl = null).
- **AI Vector Index Purge (POST /api/v1/sessions/reset-all-enrollments):** Reinitializes FAISS IndexFlatIP to empty state and flushes vectors/metadata.json.
- **Frontend Confirmation Modal:** Added dual-confirmation 'Reset All Faces' modal in students/page.tsx.

Week 10: Multi-Evidence Safe Identity Consolidation (Critical Milestone)

The Problem: Previous consolidation merged tracks purely on centroid similarity >= 0.75. Classroom empirical data showed unrelated students frequently exhibit similarity between 0.70 and 0.76, causing two different physical students to incorrectly merge into one cluster.

Multi-Evidence Mathematical Formulation:

The system now enforces a strict multi-evidence cross-sample matrix. Two candidate groups A and B are only merged if ALL conditions below are satisfied:

1. **Centroid Cosine Similarity:** $S_C(c_A, c_B) \geq 0.82$
2. **Representative Medoid Similarity:** $S_C(r_A, r_B) \geq 0.78$
3. **Multi-Sample Evidence Count:** At least 3 independent sample comparisons ≥ 0.80
4. **Top-4 Cross-Sample Mean:** mean(Top 4 Cross Sims) ≥ 0.82
5. **Anti-Contradiction Guard:** Minimum cross-similarity ≥ 0.52 and Median ≥ 0.68
6. **Anti-Chaining Transitive Integrity:** Minimum pairwise track similarity across all tracks in A and B ≥ 0.75
7. **Zero FAISS Identity Contradiction:** If group A matches Student X and group B matches Student Y with $X \neq Y$, merge is HARD REJECTED.

Core Invariant: *The system strictly prefers* ***UNDER-MERGING*** *over* ***FALSE-MERGING****. If evidence is ambiguous, identities remain separate.*

Week 11: Internal Cluster Purity Validation & Automatic Splitting

Primary Objective: Validate the internal purity of all post-clustering candidate groups and automatically split any cluster containing multiple distinct identity modes.

- **Connected-Component Graph Analysis:** Built Step 4.5 in classroom_pipeline.py. Constructs track centroid adjacency matrix at identity threshold 0.76 and runs Breadth-First Search (BFS) to identify connected components.
- **Automatic Multi-Identity Splitting:** If a cluster decomposes into > 1 component, it is automatically partitioned into separate, pure identity candidates with recomputed centroids, medoids, and quality scores.
- **Cluster Evidence & Purity Scoring:** Computes internalPurity (mean internal similarity percentage) and assigns an identityEvidence rating: **Strong** (≥6 samples, ≥2 tracks, ≥82%), **Moderate**, or **Weak**.
- **UI Indicator Badges:** Enhanced ClusterCard in face-enrollment/page.tsx with Evidence, Purity, and MULTI-ID: NO / YES badges.

Week 12: Comprehensive Automated Test Suite & Enterprise Verification

Primary Objective: Develop a 12-test automated unit test suite covering all boundary conditions of identity consolidation, and execute full production build verification.

Unit Test Verification Matrix (apps/ai-service/tests/test_identity_consolidation.py):

Test Case	Condition / Invariant Tested	Status
`test_1`	Different students (~0.73 sim) stay in 2 distinct clusters	**PASSED**
`test_2`	Single sample spike (0.81) with low remaining sim rejects merge	**PASSED**
`test_3`	High centroid with poor cross-sample comparisons rejects merge	**PASSED**
`test_4`	Multi-track same physical person merges with Strong evidence	**PASSED**
`test_5`	Frontal and angled tracks of same student merge cleanly	**PASSED**
`test_6`	Tracks split by temporary occlusion merge into 1 cluster	**PASSED**
`test_7`	Impure multi-identity cluster is auto-split into pure clusters	**PASSED**
`test_8`	Anti-chaining guard blocks non-transitive merge (A-B vs A-C)	**PASSED**
`test_9`	Contradictory enrolled FAISS student identities never merge	**PASSED**
`test_10`	Cluster purity & evidence levels calculated accurately	**PASSED**
`test_11`	Invalid/empty embedding shapes safely handled without crash	**PASSED**
`test_12`	Weak singleton glance track isolated as TENTATIVE / Weak	**PASSED**

Production Build Results: Next.js 16 Turbopack build (npm run build in apps/web) completed with exit code 0, successfully compiling all 63 static and dynamic routes.

Conclusion & Project Status

The SmartCampus AI platform is fully architected, implemented, and verified. It provides seamless contactless attendance, robust face enrollment, strict biometric privacy compliance, and mathematical guarantees against false student identity merging. The system is production-ready for classroom deployment.